

Intelligent Document Extraction, Validation & API Platform

AI-powered financial document extraction, validation, persistence and API platform

Supported Document Types:

Invoice • Balance Sheet • Profit & Loss Statement • Cash Flow Statement

Supported File Formats: PDF (Native Digital & Scanned/Image-based), JPG, PNG | Multi-page up to 3 pages per document

Submission Scope:	Working Deployed Solution + Public GitHub Repository + Documentation
Technology Stack:	FastAPI, Google Gemini 3.6 Flash (Multimodal Vision), PyMuPDF, SQLAlchemy, SQLite, Jinja2, Docker
Core Focus:	Extraction Accuracy, API Standards, Strict Mathematical Validation, Complete Test Coverage

Problem Statement & System Objectives

Platform Ingestion & Extraction Requirements

- **Dual Ingestion:** Accept financial documents through both an interactive Web Frontend and automated REST APIs.
- **Format Coverage:** Support PDF, JPG, and PNG files with up to 3 pages per document.
- **Scanned & Digital Docs:** Support native digital PDFs as well as scanned/image-based documents transparently.
- **Comprehensive Field Capture:** Extract all meaningful visible fields, tables, line items, and multi-period financial schedules.
- **Standardized Contract:** Return a deterministic, consistent, structured JSON contract with confidence & source quotes.

Financial Validation & Enterprise Governance

- **Financial Validation:** Automatically validate document-specific arithmetic (line totals, parity, profit, cash reconciliation).
- **Persistence:** Store all processed results in a queryable relational database with upsert capabilities.
- **API Retrieval & Dashboard:** Provide indexed retrieval via REST endpoints and a complete web dashboard view.
- **Live Cloud Deployment:** Containerized Docker solution deployed globally with health check monitoring.
- **Zero Hallucination Rule:**
"Missing information must be represented as null. The system must not invent or infer values."

Modular Multi-Tier Solution Architecture



Frontend Tier • HTML5, CSS3, JavaScript (Fetch API) • Drag & drop document upload • Document type selector (4 types) • Dashboard table with status badges • Comprehensive result viewer • Raw JSON view toggle	Backend Tier • FastAPI asynchronous framework • 4 modular RESTful endpoints • Automatic Swagger/OpenAPI docs • Pydantic v2 validation models • Structured logging & error handler • CORS & Multipart streaming	Processing Pipeline • File Validator: MIME, size, pages • OCR: PyMuPDF text & 200 DPI images • AI Engine: Gemini 3.6 Flash multimodal • Validation: Arithmetic reconciliation • <i>Validation runs independently from AI</i>	Persistence Tier • SQLAlchemy 2.0 ORM • Embedded SQLite database • Indexed document_name column • Full structured result_json storage • Upsert-by-name latest semantics • Automated schema initialization
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10-Step End-to-End Processing Pipeline

1. Upload

User submits document (PDF/JPG/PNG) & selects document type via Web UI or POST /api/v1/documents/process.

2. Document Validation

Validates MIME type, file integrity, readability, and enforces maximum page limit (≤ 3 pages).

3. Text Extraction / OCR

PyMuPDF extracts digital text and renders pages to 200 DPI images for multimodal inspection.

4. AI Field & Table Extraction

Gemini 3.6 Flash extracts header fields, line items, and multi-period financial tables.

5. Structured JSON Generation

Strict JSON schema enforces exact typing, null values for missing data, and evidence tracking.

6. Financial Calculation Validation

Independent validation engine verifies mathematical relationships and variances.

7. Evidence & Confidence

Extracts confidence scores (0.0–1.0), source text snippets, and page numbers per field.

8. Store Result

Saves full JSON payload and queryable metadata in SQLite database via SQLAlchemy.

9. PASS / FAILED Status

Assigns overall document status (PASS if readable and extracted, FAILED on corruption/format error).

10. Dashboard & API Delivery

Returns response instantly to caller and updates dashboard table with latest record.

Supported Financial Documents & Extraction Schema

Document Type	Extracted Fields & Structural Components	Mandatory Financial Validation Checks
Invoice Retail Receipts & B2B Tax Invoices	Invoice number, date, vendor name/address/tax ID, customer name/address/tax ID, currency, subtotal, tax rate, tax amount, discount, total amount, cash paid, change, rounding adjustment, and line items (Qty, Unit Price, Amount, VAT%).	• $\text{Quantity} \times \text{Unit Price} \approx \text{Item Amount}$ • $\text{sum}(\text{line_item_amounts}) \approx \text{Subtotal}$ • $\text{Subtotal} + \text{Tax} - \text{Discount} + \text{Rounding} \approx \text{Total}$ • $\text{Cash Paid} - \text{Total Amount} \approx \text{Change}$
Balance Sheet Consolidated Bank Statements	Company name, report title, as-of date, currency, reporting unit, auditor notes, and comparative periods containing Capital & Liabilities items + schedule refs and Asset items + schedule refs.	• Assets $\approx$ Liabilities + Equity (Total Assets $\approx$ Total Capital & Liabilities) • $\text{sum}(\text{Asset Components}) \approx \text{Total Assets}$ • $\text{sum}(\text{C\&L Components}) \approx \text{Total C\&L}$
Profit & Loss Income Statement	Company name, period, currency, revenue schedules (Interest earned, Other income), expenditures (Interest expended, Operating expenses, Provisions), net profit before/after minority interest, appropriations.	• $\text{sum}(\text{Income items}) \approx \text{Total Income}$ • $\text{sum}(\text{Expenditure items}) \approx \text{Total Expenditure}$ • $\text{Total Income} - \text{Total Expenditure} \approx \text{Net Profit}$ • Group Profit & Appropriation Balance checks
Cash Flow Statement Multi-Page Statements	Operating activities, investing activities, financing activities, foreign exchange translation adjustments, net increase/decrease in cash, opening cash, cash acquired on amalgamation, closing cash balance.	• $\text{Operating} + \text{Investing} + \text{Financing} + \text{FX} \approx \text{Net Change in Cash}$ • $\text{Opening Cash} + \text{Net Change} + \text{Adjustments} \approx \text{Closing Cash}$

REST API Design & JSON Response Contract

API Endpoints Specification

POST /api/v1/documents/process

Uploads & processes financial document. Form: file (binary) + document_type (invoice, balance_sheet, profit_and_loss, cash_flow_statement).

GET /api/v1/documents/{document_name}

Retrieves the latest structured result, extraction data, and validation audits for a specific document name.

GET /api/v1/documents

Returns summary index of all processed documents (status, confidence, timestamp) to populate dashboard.

GET /api/v1/health

Operational health check probe returning service status.

Interactive OpenAPI Documentation:

FastAPI provides automatic interactive Swagger documentation at /docs and ReDoc at /redoc.

Standard Structured JSON Contract

```
{
  "document_name": "invoice_101.pdf",
  "document_type": "invoice",
  "processing_status": "PASS",
  "overall_confidence": 0.98,
  "file_validation": {
    "file_type": "application/pdf",
    "is_supported": true, "is_readable": true,
    "page_count": 1, "status": "PASS"
  },
  "extracted_data": {
    "invoice_number": {"value": "INV-2024", "confidence": 0.99},
    "total_amount": {"value": 138.90, "confidence": 0.98}
  },
  "validation": {
    "checks": [{"name": "total_check", "status": "PASS"}],
    "overall_status": "PASS", "issues": []
  },
  "processing_metadata": {"ocr_used": false, "time_ms": 1420}
}
```

Financial Validation Engine & Status Evaluation

Independence of Extraction & Validation

The financial validation engine runs as an independent deterministic verification layer after extraction. AI extracts values; Python arithmetic reconciles formulas.

Validation Check Schema

Every check outputs an audit record containing:

- **name:** Check identifier (e.g. invoice_total_check)
- **formula:** Human-readable formula audited
- **operands:** Input variables evaluated
- **calculated_value:** Result computed by Python
- **reported_value:** Number printed on document
- **variance:** Absolute difference |calculated - reported|
- **status:** PASS | FAIL | NOT_APPLICABLE

Status Rules & Tolerance Handling

PASS Criteria:

Variance is within the allowed rounding tolerance margin (± 1.0). Handles minor floating-point or currency rounding differences.

FAIL Criteria:

Variance exceeds allowed tolerance, or line totals do not reconcile with subtotal. Highlighted in red in the UI.

NOT_APPLICABLE Rule:

If a required field is not present in the document (e.g. cash paid on a credit invoice), the check returns **NOT_APPLICABLE**.
Missing fields are never guessed, inferred, or assumed.

Core Verification Formulas: **[Invoice]:** $Qty \times Price \approx Amount$ | $sum(Lines) \approx Subtotal$ | $Subtotal + Tax - Disc \approx Total$ | $Cash - Total \approx Change$
[Balance Sheet]: $Total\ Assets \approx Total\ Capital\ \&\ Liabilities$ | **[P&L]:** $Income - Expenses \approx Net\ Profit$ | **[Cash Flow]:** $Op + Inv + Fin + FX \approx Net\ Change$ | $Open + Change \approx Close$

Frontend Dashboard, Persistence & Cloud Deployment

Frontend Web Dashboard

- **Document Ingestion:** Dropdown document type selection + drag-and-drop file upload.
- **Execution Feedback:** Animated progress bar and status indicator during processing.
- **Documents Table:** View all processed files, document types, status badges, confidence %, and timestamps.
- **Result Detail View:** Clean key-value grid for extracted fields with confidence color badges (High/Med/Low).
- **Table Visualizer:** Tabular line-item breakdown & comparative period schedules.
- **Audit Breakdown:** Visual validation checks table highlighting PASS / FAIL / NA.
- **Raw JSON Toggle:** Formatted JSON view for debugging.

Persistence, Cloud & Security

Relational Persistence:

- SQLite relational database managed via SQLAlchemy ORM.
- Stores document_name, document_type, status, confidence, and complete result JSON payload.
- Upsert semantics: re-processing same document updates record.

Production Cloud Deployment:

- **Dockerized:** Hermetic python:3.12-slim container.
- **Render Cloud:** Auto-deployed web service with dynamic port binding (\$PORT).
- **Public URL:** Globally accessible frontend and REST API.

Security Architecture:

- API keys strictly managed via environment variables (.env).
- .gitignore protects keys and local database files from GitHub.
- Controlled error handling: zero raw stack traces exposed.

Testing Suite & Software Engineering Rigor

Test Suite Category	Automated Test Scenarios Evaluated	Engineering Outcome
API Processing Flow test_api.py (9 tests)	• Health check endpoint (200 OK) • Document listing index endpoint • Unsupported file extension rejection (400 Bad Request) • Invalid document_type rejection (422 Unprocessable Entity) • Empty file rejection (400 Bad Request) • Corrupted PDF handling (graceful FAILED status) • Non-existent document 404 handling • Dashboard & detail page HTML rendering	100% Pass Rate Rigorous HTTP status code compliance and exception isolation.
Financial Calculations test_validation.py (13 tests)	• Invoice line item multiplication (Qty × Price) • Line item subtotal summation check • Grand total tax & discount calculation (Pass & Fail cases) • Cash & change point-of-sale math • Balance sheet Assets = Liabilities parity • P&L income & expenditure component sums and net profit derivation • Cash flow activity sums and opening→closing cash reconciliation • NOT_APPLICABLE status handling for omitted optional fields	100% Pass Rate Accurate validation across all 4 financial document accounting rules.
Normalization Helpers test_extraction.py (15 tests)	• safe_float parenthesized negatives: (500.00) → -500.00 • String numbers with comma formatting: "1,234,567.89" → 1234567.89 • None and malformed string handling (clean null conversion) • get_field_value: extraction of scalars and nested value dictionaries • get_nested_value: deep path traversal without KeyError exceptions	100% Pass Rate Zero runtime crashes from unexpected OCR character formatting.

Total Test Suite: 37 passing unit & integration tests executed via pytest with automated coverage across all layers.

Current Limitations & Production Improvements

Current Technical Limitations

- **Scan Quality Sensitivity:** Highly degraded, blurred, crumpled, or low-DPI scans can impact decimal place readability.
- **Complex Non-Standard Layouts:** Heavily customized multi-column banking footnotes or unconventional tables can require specialized post-processing.
- **LLM Inference Latency & Quotas:** Cloud multimodal LLM calls take several seconds per document and require API rate-limit management on free-tier quotas.
- **Ephemeral Local Storage:** SQLite embedded storage on free-tier containers does not persist across container re-deployments.

Enterprise Production Roadmap

- **Managed Relational Database:** Transition to AWS RDS / Google Cloud SQL PostgreSQL with JSONB indexing for enterprise durability.
- **Durable Object Storage:** Store uploaded files and rendered page images in AWS S3 or Google Cloud Storage with presigned access.
- **Asynchronous Task Queue:** Integrate Celery + Redis for async document processing, decoupling ingestion from AI extraction.
- **Security & Access Control:** Implement OAuth2 / JWT authentication, role-based access control (RBAC), and rate limiting.
- **Human-in-the-Loop (HITL):** Automated routing of low-confidence extractions (< 80%) to human verification portals.
- **Observability:** Structured OpenTelemetry distributed tracing, Prometheus metrics, and Sentry error alerting.

Live Demonstration & Evaluator Walkthrough

Step 1	Open Deployed Frontend — Access the web application dashboard in any browser at the public deployment URL.
Step 2	Select Document Type — Choose one of: Invoice, Balance Sheet, Profit & Loss, or Cash Flow Statement.
Step 3	Upload Document — Attach a sample file (e.g. receipt JPG or multi-page financial PDF) and click 'Process'.
Step 4	Inspect Extracted Data — Review extracted key-value fields, confidence badges, and multi-row line item tables.
Step 5	Verify Financial Audits — Examine mathematical validation checks: formulas, calculated vs reported, variances, and PASS/FAIL/NA.
Step 6	Inspect Raw JSON Contract — Click 'Toggle Raw JSON' to review the standardized API payload structure and metadata.
Step 7	Explore Swagger & REST API — Navigate to /docs, execute GET /api/v1/documents, and test GET /api/v1/health.

Evaluation Principle: The system runs transparently in front of evaluators. Upload any provided benchmark file to inspect real-time OCR, extraction, validation, and database indexing.

Comprehensive Case Study Deliverables Checklist

Core Requirements Verification

- ✓ **Public GitHub Repository:** Clean commit history, modular structure.
- ✓ **Live Deployed Frontend:** Accessible web dashboard on Render cloud.
- ✓ **Live Backend API:** Fully functional RESTful endpoints with CORS.
- ✓ **Interactive Swagger Docs:** OpenAPI specification available at /docs.
- ✓ **Health Check Endpoint:** Monitoring probe at /api/v1/health.
- ✓ **Comprehensive README:** Setup guide, architecture, API docs & trade-offs.
- ✓ **Solution Presentation PDF:** Professional 12-slide technical submission.
- ✓ **Sample JSON Outputs:** Real extracted outputs in sample_outputs/.

Technical Coverage Verification

- ✓ **All 4 Document Types:** Invoice, Balance Sheet, P&L, and Cash Flow.
- ✓ **Scanned Document Support:** Tested on scanned banking & receipt images.
- ✓ **Relational Persistence:** SQLite storage with SQLAlchemy ORM upsert.
- ✓ **Automated Test Suite:** 37 passing pytest unit and integration tests.
- ✓ **Validation Failure Testing:** Real-world discrepancy detection tested.
- ✓ **Security & Error Handling:** Zero exposed API keys or raw stack traces.
- ✓ **Engineering Discipline:** Clean design patterns, no hardcoded extraction.
- ✓ **Production Roadmap:** Clear articulation of limits and enterprise scale.

Status: Deployed, testable, documented and ready for technical walkthrough.